

Taiwan Experience Education Program (TEEP): Research, Technology, Culture, and Life in Taiwan

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1. Introduction

This report presents my experience participating in the Taiwan Experience Education Program (TEEP), focusing on my research internship and my experiences living and exploring Taiwan. The main focus of my internship was the development and evaluation of a Robot Nurse Helper using the Kebbi social robot, integrating bilingual interaction, and healthcare applications. The project included functions related to cataract surgery and object detection, as well as testing in a hospital environment, where we received valuable feedback for further improvement. In addition to research, I experienced Taiwan's academic and technological environment through activities such as the AIWAVE exhibition and a visit to Taipei 101. I also explored various cultural and natural destinations, including Qingtiangang, Jiufen, Shifen, Golden Waterfall, Tamsui, and Beitou. These experiences contributed to my technical knowledge, research skills, cultural understanding, and personal growth during the TEEP program.

2. Research Project: Robot Nurse Helper Using Kebbi

The main focus of my internship was the development of a Robot Nurse Helper using the Kebbi robot. The project aimed to explore how artificial intelligence and robotics could be applied in a healthcare environment to provide useful assistance and improve interaction with patients. The project combined robotics, artificial intelligence, computer vision, and human-robot interaction.

During the internship, I worked on several important features of the system. The robot was developed with both English and Chinese interaction, making it more suitable for use in Taiwan while also supporting international users. We also worked on functions related to cataract surgery and its queries, allowing the robot to provide relevant information and assistance during the healthcare process. Another important component was object detection, which allowed the robot to recognize objects in its surroundings and demonstrated the potential in a healthcare robotic system.

Working on the project gave me practical experience in integrating different technologies into a single robotic system. It also helped me understand that developing healthcare technology requires more than technical implementation. The system must be understandable, practical, and comfortable for the people who will interact with it.

3. Hospital Testing and Research Experience

One of the most valuable parts of my internship was testing the Kebbi robot in a real hospital environment. Before the hospital evaluation, much of the development and testing was

performed in a controlled environment. Hospital testing allowed us to see how the robot performed when interacting in a real healthcare setting.

The testing provided positive and encouraging feedback while also revealing areas where the system could be improved. This experience showed me the importance of user feedback and real-world evaluation. Some features that appear effective during laboratory testing may need adjustment when used by real patients or in a busy clinical environment.

The feedback from the evaluation helped us identify improvements in the robot's interaction and overall system performance. This iterative process of development, testing, feedback, and improvement became an important part of my research experience.

Working on this project also allowed me to collaborate with professors and students from different backgrounds. I learned how interdisciplinary teamwork can combine healthcare knowledge with artificial intelligence and robotics to address real-world problems. The experience improved my technical skills as well as my communication, teamwork, problem-solving, and research abilities.



4. Academic and Technology Experience in Taiwan

TEEP also gave me the opportunity to experience a different academic and technological environment. Working with professors and students in Taiwan helped me understand different approaches to research, communication, teamwork, and project development. Being part of an international environment also improved my confidence in communicating and collaborating with people from different backgrounds.

One of the most interesting academic experiences was visiting the AIWAVE exhibition. As an artificial intelligence student working on a robotics project, the exhibition was particularly meaningful to me. I was able to observe new technologies and different applications of artificial intelligence and gain a broader understanding of how AI is being developed for practical use. Seeing technologies beyond the university laboratory encouraged me to think about possible future applications of AI, and robotics.

We also had the opportunity to visit Taipei 101 through an activity organized with our professor and the host institution. Besides being a memorable cultural experience, it provided

an opportunity to spend time with other students outside the research environment and learn more about Taiwan.

These experiences helped me understand that an internship is not only about completing a project but also about learning from the surrounding academic and technological environment.



5. Life and Travel in Taiwan

Living in Taiwan was one of the most enjoyable parts of my TEEP experience. I had the opportunity to visit Longshan Temple, where I learned more about Taiwanese traditions, religious practices, and local culture. I also found the transportation system extremely convenient, especially the MRT and buses, which made it easy to travel around and explore different places. I found Taiwan to be a peaceful and welcoming country, and I enjoyed the experience of living here and discovering something new almost every day. These everyday experiences helped me feel more comfortable in a new environment and gave me a deeper appreciation of Taiwanese culture and lifestyle.

Outside the university, I visited many natural, historical, and cultural destinations. Qingtiangang Grassland was one of the memorable natural places I visited, with its open grassland and mountain scenery providing a relaxing break from the research environment. I also visited Jiufen Old Street, where I experienced traditional streets, local food, shops, and the unique atmosphere of the historic mountain town.

Another memorable trip was to Shifen Waterfall, where I experienced one of Taiwan's well-known natural attractions. I also visited Golden Waterfall, which provided a different landscape and showed the natural diversity of northern Taiwan.

In addition, I explored Tamsui, enjoying its riverside environment, local streets, and cultural atmosphere, and visited Beitou, which provided an opportunity to experience another part of Taiwan's natural and cultural heritage. Along with these destinations, I visited several other places during my stay, each providing a different perspective on Taiwanese life.

Traveling around Taiwan helped me become more independent and confident. Planning trips, using public transportation, managing time, communicating in an unfamiliar environment,

and adapting when plans changed were all valuable experiences that extended beyond tourism.



6. Challenges, Learning, and Personal Growth

Like any international experience, my internship also involved challenges. Adapting to a new country, academic environment, language, transportation system, and daily routine required flexibility and patience. The research project also involved technical challenges, particularly when integrating different functions and testing the robot in a real-world healthcare environment.

However, these challenges became important learning opportunities. I learned that research requires continuous improvement and that unexpected problems are a natural part of development. Hospital testing particularly taught me the importance of designing technology around real users rather than focusing only on whether the technical system works.

The experience also helped me become more independent and adaptable. Living in another country encouraged me to step outside my comfort zone, communicate with new people, and manage unfamiliar situations. At the same time, working with an international research team improved my confidence and strengthened my ability to collaborate.

The combination of research, technology exposure, travel, and cultural experiences gave me a broader perspective than I could have gained through classroom learning alone.

7. Conclusion

The Taiwan Experience Education Program was a valuable combination of research, technical development, practical experience, cultural exchange, and personal growth. Through the Kebbi Robot Nurse Helper project, I gained practical experience in artificial intelligence, robotics, bilingual interaction, object detection, healthcare applications, system integration, and real-world testing. The hospital evaluation was particularly meaningful because it allowed us to receive practical feedback and improve the system.

Beyond the research project, visiting the AIWAVE exhibition and Taipei 101 expanded my understanding of Taiwan's technology and culture. Exploring places such as Longshan temple,

Qingtiangang, Jiufen, Shifen, Golden Waterfall, Tamsui, Beitou, and other destinations allowed me to experience Taiwan beyond the academic environment.

Overall, I feel that it was an experience that helped me grow both academically and personally. Working on the Kebbi Robot Nurse Helper allowed me to apply my technical knowledge to a real healthcare project and understand the importance of testing and improving a system through practical feedback. At the same time, living in Taiwan gave me the chance to experience a different culture, meet new people, and visit many memorable places, which made my stay even more meaningful.

I sincerely appreciate the guidance and support provided by my professor and Chang Gung University throughout the program, as well as the opportunity made possible through TEEP. The knowledge, friendships, cultural experiences, and confidence I gained during my time in Taiwan will continue to influence my future studies and career.